Response to Reviewers

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Title: Objective-Dependent Active Learning and Calibrated Uncertainty for Sample-Efficient Discovery of Rare-Earth Extractants: A Benchmark on 1,202 Experimental Distribution Coefficients with Prospective Validation.

We thank the reviewer for a careful and constructive report. Every point has been addressed with **new experiments on the real data**; no claim is asserted without a computed result. The revision is substantial: the manuscript now reports **five findings** (adding a leakage-controlled generalisation analysis and a definition-dependent discovery analysis), **four new tables**, and **five new figures**, all regenerated by released scripts. Below, each comment is quoted and followed by our response; section, table, and figure numbers refer to the revised manuscript.

Summary of major changes

1. **Leakage-controlled splits** (Fig. 1, Table 1): row-level CV is shown to be strongly optimistic; ligand-, source-, and family-disjoint splits collapse R^2 from 0.88 to $\sim$0.2 or below. Reframes finding (i).
2. **Ligand-batch acquisition** (Fig. 4): a realistic decision unit (select a ligand, measure all its conditions); the objective-dependent ordering persists.
3. **Definition of a "top" system** (Fig. 5, Table 2): recall for rows, ligand–metal pairs, and unique ligands; the strategy ranking inverts.
4. **Subgroup-resolved calibration** (Fig. 6, Table 3): ligand-disjoint train/calibration/test; coverage by $\log D$ region and lanthanide; Mondrian conformal; and a demonstration that non-monotone calibration reorders the ranking.
5. **Broader model/UQ baselines** (Fig. 7, Table 4): random forest, quantile regression forest, Gaussian process, gradient boosting, MC-dropout Bayesian NN, with uncertainty-ranking-quality and discovery metrics.
6. **Expanded literature**: 17 verified references (2013–2025).

Reviewer 1 (publish after major revisions)

We are grateful for the positive assessment of the objective-dependent framing, the random-forest baseline, the multi-seed curves, the independent-dataset replication, and the prospective test.

1.1 Split rigor / data leakage

If train/test splits are made at the row level, closely related measurements from the same ligand or condition family can appear in both training and testing...a more stringent split by ligand scaffold, publication source, or ligand family would better represent discovery of genuinely new extractants.

Response. We agree, and this is now a central result. The released matrix has no SMILES, but the grouping structure is recoverable: identical ECFP bit-vectors identify the 93 ligands, the metal-descriptor block the 14 lanthanides, and the citation index the 45 sources. Under leakage-controlled splits (new §"Realistic generalisation…", Fig. 1, Table 1):

Split	RF R^2
Random (row-level)	0.88 ± 0.03
Ligand-disjoint	0.20 ± 0.31
Source-disjoint	-0.03 ± 0.80
Ligand-family (ECFP-Tanimoto, scaffold proxy)	-0.24 ± 0.50
Prospective (4 new ligands)	0.70

The row-level number is strongly optimistic. We now state that in-distribution R^2 is *not* a deployability estimate, and that the prospective $R^2 = 0.70$ sits between the extremes because the four new ligands are close analogues of training chemotypes. We cite Sheridan 2013 and Guo et al. 2024/2025. Because SMILES are unavailable, we are transparent (Limitations) that the family split is an ECFP-Tanimoto proxy for a scaffold split.

1.2 Experimental decision unit / batch acquisition

Selecting one ligand can imply measuring multiple lanthanides, acidities, diluents, or concentrations…a ligand batch acquisition simulation…would make the benchmark more chemically realistic.

Response. Added (new §, Fig. 4). We re-ran the benchmark with a ligand-batch unit: selecting a ligand reveals all its measurements (median 14), and the test set is ligand-disjoint. The design-side dichotomy is clear: exploitation gives the best top-10 ligand recall (1.00) while random/exploration lags (0.83). On prediction, all R^2 are low under this hard test, with random best (0.30) by a small margin over exploitation (0.29). We report this honestly: the ordering is preserved, though the prediction-side gap narrows once train–test ligand overlap is forbidden.

1.3 Definition of "top-10 recall"

The 'top 10 recall' metric should specify whether 'top' systems are unique ligands, ligand–metal pairs, or condition specific measurements.

Response. Added (new §, Fig. 5, Table 2). Recall is reported under all three definitions, and the ranking inverts: exploitation dominates at the row level (greedy 1.00 vs random 0.45), but at the ligand level broad sampling recovers most top ligands (random 0.95 vs greedy 0.82; EI best at 0.97). We caution that an unstated "top-k recall" can reverse the practical conclusion.

1.4 Conformal rigor and subgroup coverage

The conformal procedure needs more precise reporting…reporting coverage by chemical subgroup, logD range, lanthanide identity, and prospective ligands…non monotonic or locally adaptive calibration could alter rankings.

Response. Fully reworked (new §, Fig. 6, Table 3). (a) Train/calibration/test are now ligand-disjoint. (b) Coverage of nominal-80% intervals is reported by subgroup: raw intervals under-cover (0.72), a global conformal factor restores average coverage (0.90) but leaves the high-log D region uneven, and Mondrian (group-conditional) conformal (0.87) evens the subgroups; per-lanthanide-class coverage is 0.89–0.91, prospective 0.88. (c) We quantify the non-monotone caveat: Mondrian reorders the uncertainty ranking versus the global factor (Spearman $\rho = 0.85 < 1$), so locally adaptive calibration *can* change acquisition even though a rank-preserving global factor cannot. We retain the in-distribution result (calibration error $0.103 \rightarrow 0.013$) and frame the two as complementary.

1.5 Broader baselines and metrics

Gaussian process regression, quantile random forests, gradient boosting, Bayesian neural networks…Reporting top-k enrichment, regret, hit rate…and Pareto performance…would strengthen the connection to separations chemistry.

Response. Added a five-estimator comparison (new §, Fig. 7, Table 4): random forest, quantile regression forest, Gaussian process (PCA-reduced), gradient boosting, and an MC-dropout Bayesian NN, each with a split-conformal wrapper, on a ligand-disjoint split, scored on predictive accuracy, conformal coverage, and uncertainty-ranking quality (Spearman of σ vs. error; selective-prediction RMSE-AUC). We added the requested discovery metrics (enrichment factor, simple regret, hit-rate above $\log D \geq 1$) to Table 2 and the R^2-vs-recall Pareto view (Figs. 3–5). No estimator uniformly dominates (Hirschfeld 2020, Scalia 2020), with high fold variance; gradient boosting is most consistent (best mean ranking quality $\rho = 0.28$) and the Gaussian process is weakest. The dichotomy and calibration-vs-acquisition distinction hold across estimators.

1.6 / 1.7 More figures and tables

Response. The revision adds five figures (splits, ligand-batch, recall definitions, subgroup calibration, model/UQ comparison) and four tables, in addition to the original three figures, all auto-generated from the results files.

1.8 More literature / references not up to date

Response. The introduction and discussion now cite 17 added, verified references (2013–2025): pool-based/batch AL (Graff 2021, Rohr 2020, Reker 2020, Bayley 2024, Bailey 2023), explore/exploit (De Ath 2021), realistic splitting (Sheridan 2013, Guo 2024/2025), group-conditional conformal (Sun 2017), early-recognition metrics (Truchon & Bayly 2007, Ash 2022), UQ comparisons (Hirschfeld 2020, Scalia 2020, Soleimany 2021, Busk 2022, Gal 2016), and current rare-earth $\log D$ ML (Udawattha & Alam 2025, Chaube 2020, Edaugal 2025).

Additional questions

The new leakage-controlled evaluation, ligand-batch acquisition, subgroup calibration, and multi-estimator comparison address the technical-quality and conclusions-supported concerns by testing every claim under realistic, non-leaking conditions and across models; the reference list is now current. We hope the reviewer finds the revised manuscript substantially strengthened.